Week 7, homework 1: SIR & SEIR Epidemic modeling

part B SIR Model

I expect infections to increase at first and peak between days 40 and 60 with around 450-550 people infected at the same time. my reasoning is that the transmission rate is initially greater than recovery rate so infections will increase while there are still enough susceptible people. as more people become infected and recover, fewer susceptible people remain, and eventually recoveries should outnumber new infections. since recovered subjects don't become susceptible again, I expect one main wave of infection.

part D SEIR Model

I expect more than one infection wave over the 1500 days although I think later waves may be smaller. SEIR includes an exposed numerical compartment to the function, meaning that subjects don't become infectious immediately after infection. It also accounts for births and deaths. I think the continous addition of susceptible subjects could allow infections to rise again, rather than disappearing entirely. I'm curious to see whether the delay introduced by the exposed stage makes these waves more spread out.

Part 2 Sensitivity Analysis

I hypothesize that the transmission rate β to have a greater effect on the infection peak compared to the recovery rate γ. I think increasing the transmission could cause infections to grow faster than the population can recover, especially early in the outbreak. So changing how quickly subjects become infected will have a stronger effect on peak.